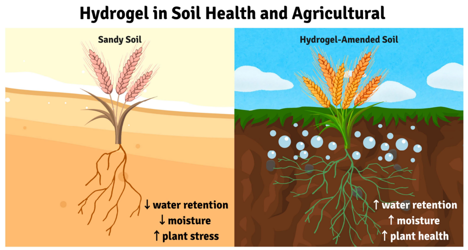
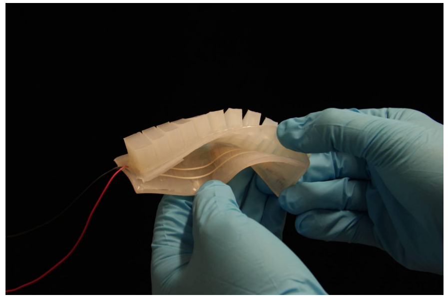
Hydrogels

Applications

Hydrogel property	Representative applications
High water content	Artificial tissues, contact lenses
Biocompatibility	Tissue engineering, wound dressings
Tunable mesh size	Drug delivery
Stimuli responsiveness	Actuators, sensors
High swelling capacity	Agriculture, absorbents
Soft mechanics	Soft robotics
Permeability	Cell culture, bioreactors





Crosslinked Polymers

Microgels

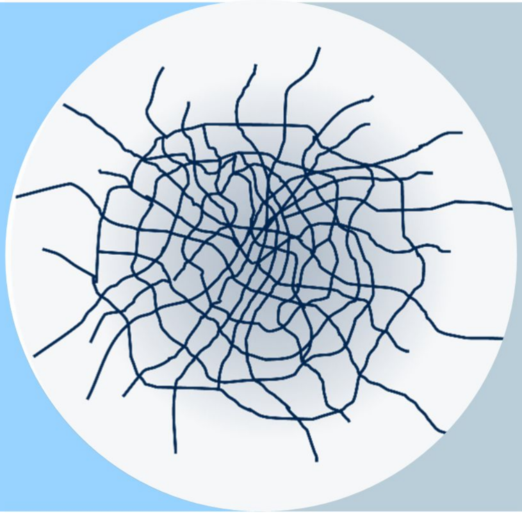
Microgels

Historical Perspective

POLYMER CHEMISTRY

crosslinked latex particles

emulsion/dispersion polymerization



COLLOID SCIENCE

soft particles

size, deformability, and interactions differing from hard particles (Si, PS)

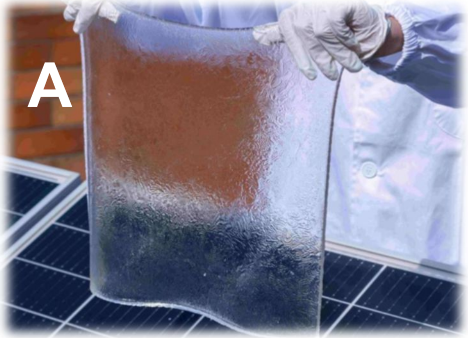
1949 Publication “Microgel, a new macromolecule” by Baker; “microgel” = cross-linked polybutadiene latex particles (*Ind. Eng. Chem.* **1949**, *41*, 511)

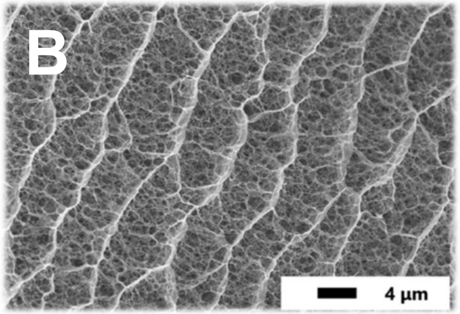
1986 Publication „Preparation of Aqueous Latices with *N*-Isopropylacrylamide“ by Robert H. Pelton (*Colloids Surf.* **1986**, *20*, 247)

Microgels

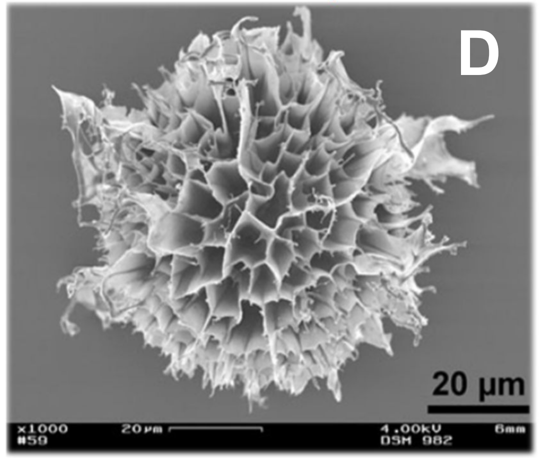
From Bulk Hydrogels to Microgels

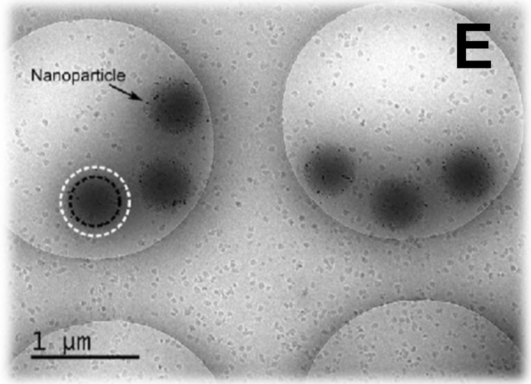
macroscopic





microscopic





A https://www.pv-magazine.com/2026/01/28/new-hydrogel-layer-tackles-solar-module-hotspots/
B *High Pressure Research* **2021**, *41*, 97
C https://ece.engin.umich.edu/stories/new-biodegradable-hydrogel-offers-eco-friendly-alternative-to-synthetics
D *Microgels by Precipitation Polymerization: Synthesis, Characterization, and Functionalization.* In: Pich, A., Richtering, W. (eds) Chemical Design of Responsive Microgels. Advances in Polymer Science, vol 234. Springer, Berlin, Heidelberg
E *ACS Omega* **2019**, *4*, 3, 4636

Microgels

What is a Microgel?

latex particle		***microgel particle***
	Diameter ≈ 100 nm–100 µm	
hard		soft
non-swelling		swelling
fixed diameter		variable diameter
(generally) rigid		deformable

Microgels

Internal Structure of a Microgel

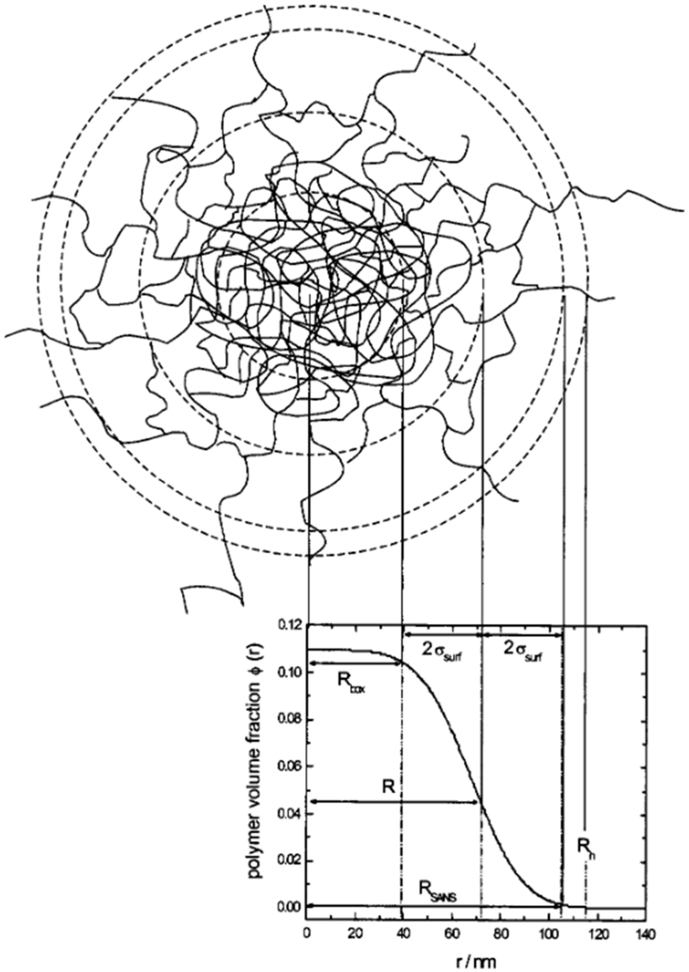


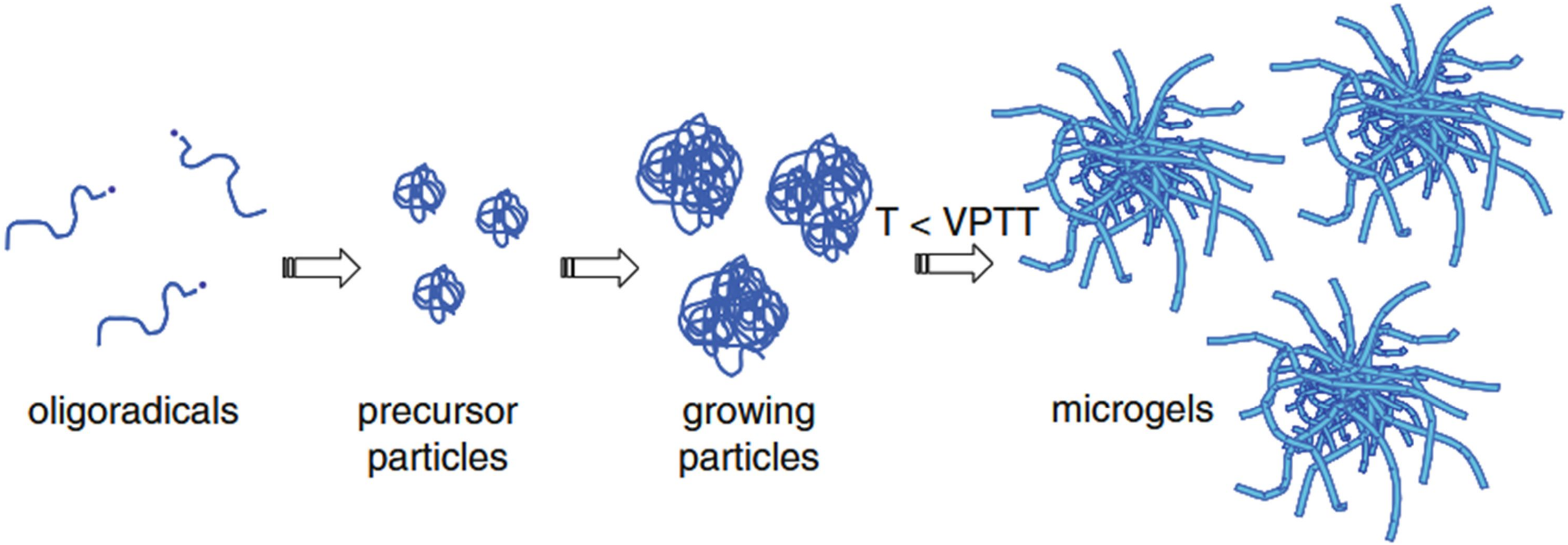
FIG. 1. Structure of PNiPAM microgels. A highly cross-linked core is characterized by a radial box profile up to $R_{box}=R-2\sigma_{surf}$. The cross-linking density decreases with increasing distance to the core described by σ_{surf}. At R the profile has decreased to half the core density. The overall size obtained by SANS is approximately given by $R_{SANS}=R+2\sigma_{surf}$ where the profile approaches zero. R_{SANS} is slightly smaller than the hydrodynamic radius R_h obtained by DLS.

J. Chem. Phys. **2004**, *120*, 6197

Microgels

Synthesis of Microgels

Precipitation Polymerization



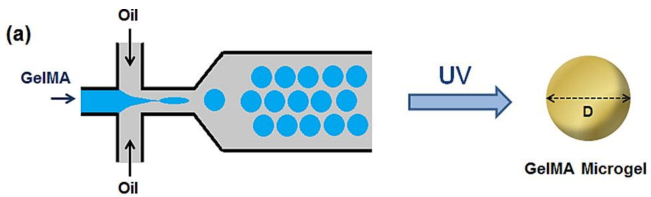
Microgels by Precipitation Polymerization: Synthesis, Characterization, and Functionalization.
In: Pich, A., Richtering, W. (eds) Chemical Design of Responsive Microgels. Advances in Polymer Science, vol 234. Springer, Berlin, Heidelberg

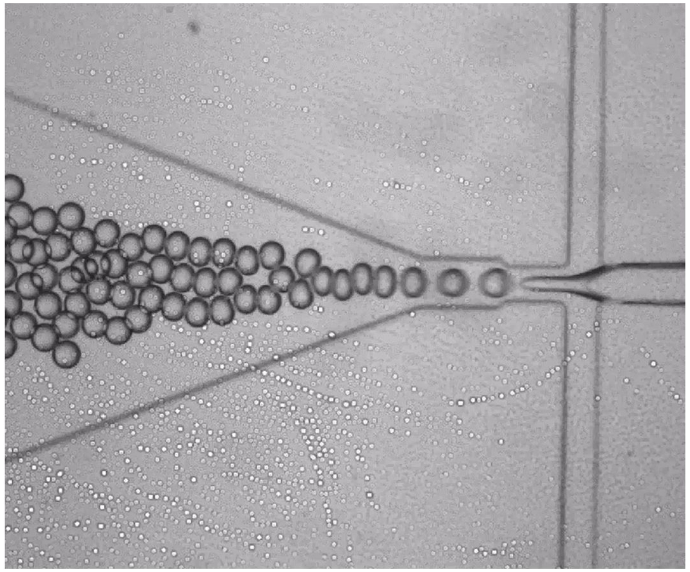
Microgels

Synthesis of Microgels

Further Methods – Towards Homogeneous Microgels

1. Controlled/living radical polymerization (most promising)
2. Continuous (starved) crosslinker addition
3. Post-crosslinking of preformed particles
4. Microfluidic synthesis
5. Seeded growth



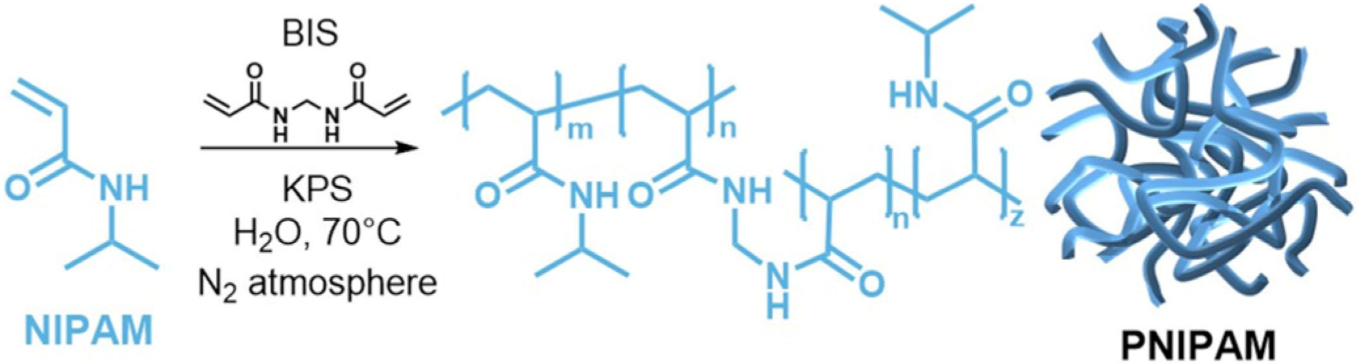


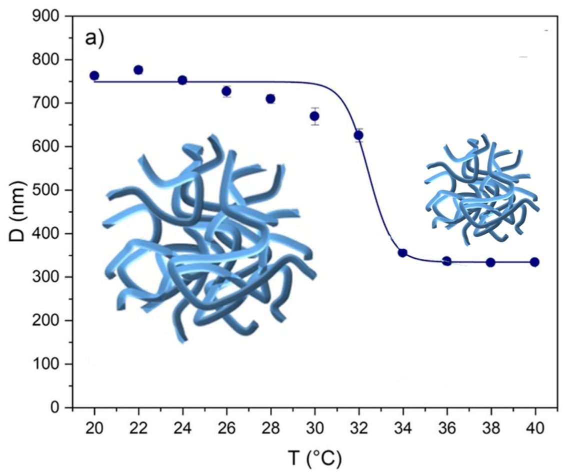
Can one obtain a *perfectly* homogeneous microgel? Probably not.

Microgels

PNIPAM Microgels

The Model System



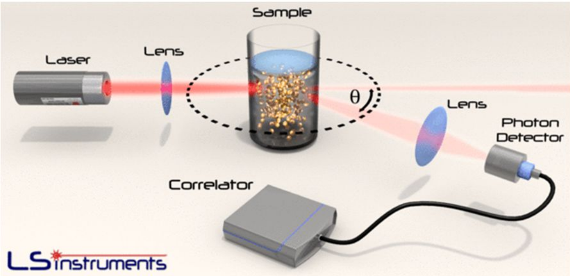


Gels **2024**, *10*, 411

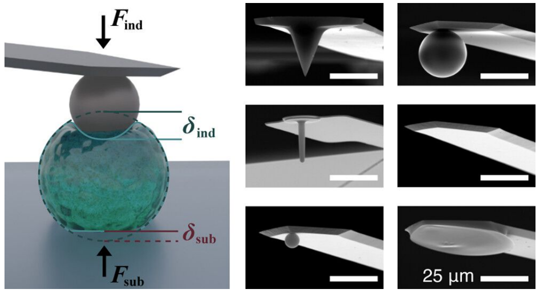
Microgels

Characterization of Microgels

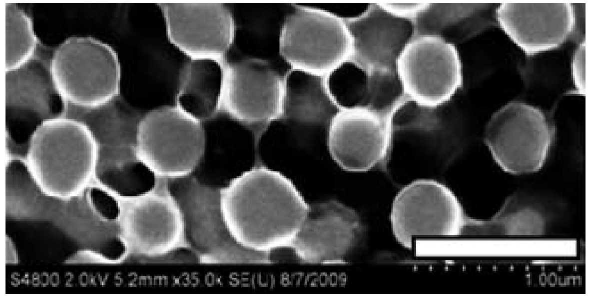
DLS
dynamic light scattering



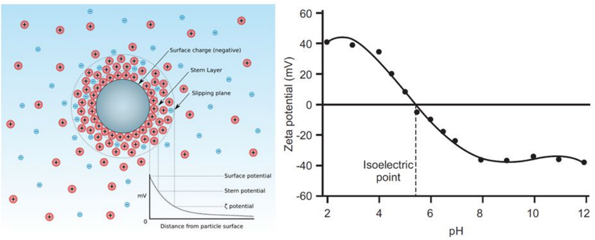
AFM
atom force microscopy



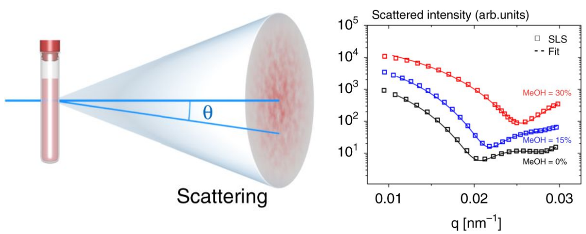
SEM
scanning electron microscopy



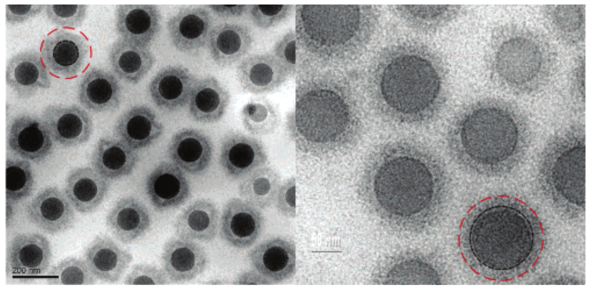
zeta potential
electrophoretic mobility



SAXS/SANS
small-angle X-ray/neutron scattering

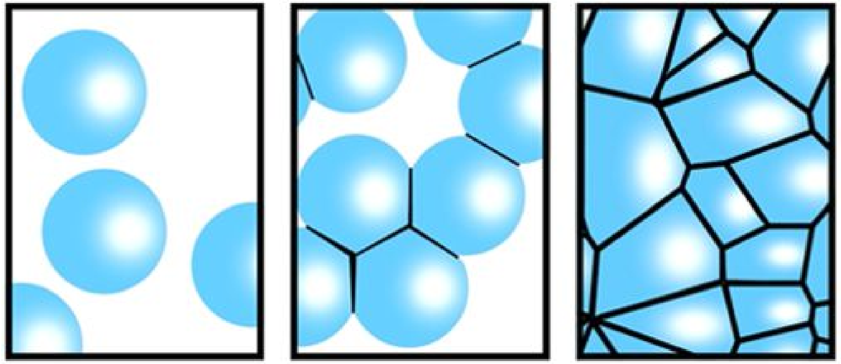


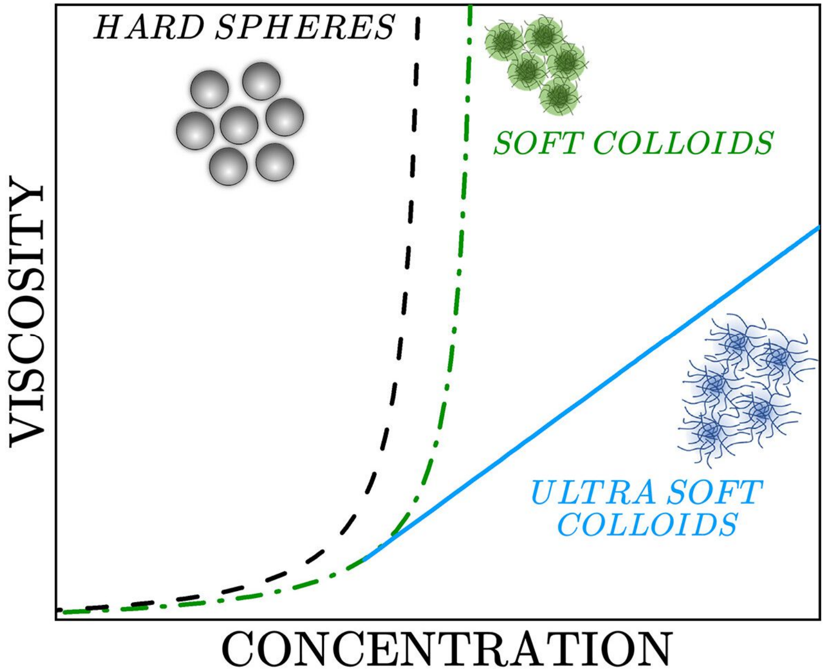
Cryo-TEM
cryogenic transmission EM



Microgels

Microgels as Soft Colloids





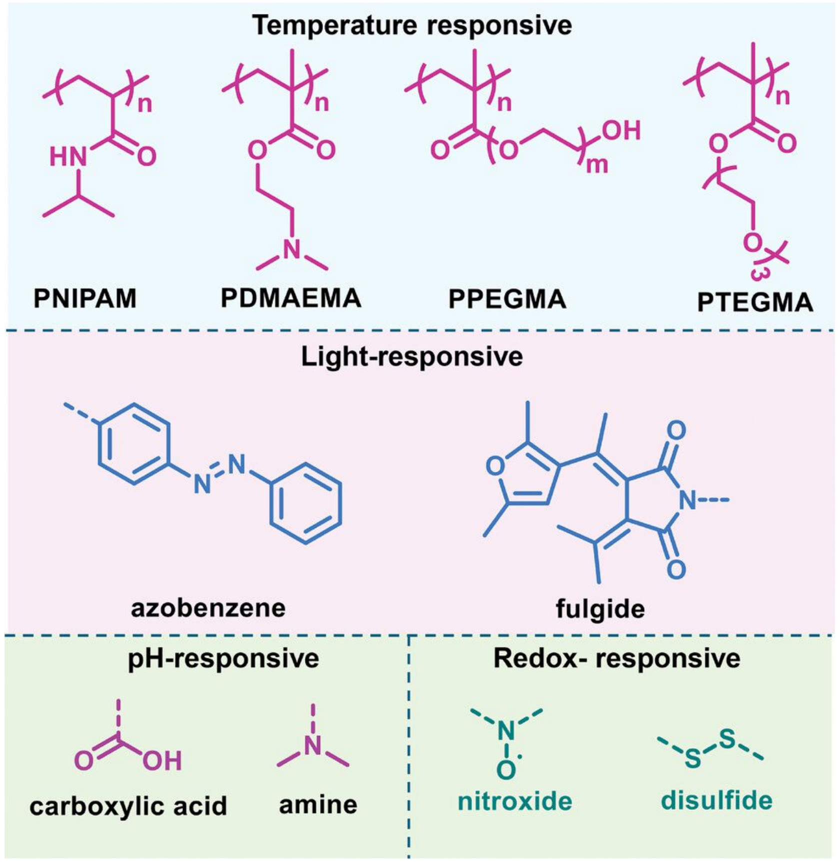
Nat. Commun. **2020**, *11*, 4315

https://onlinelibrary.wiley.com/doi/full/10.1002/pola.70233

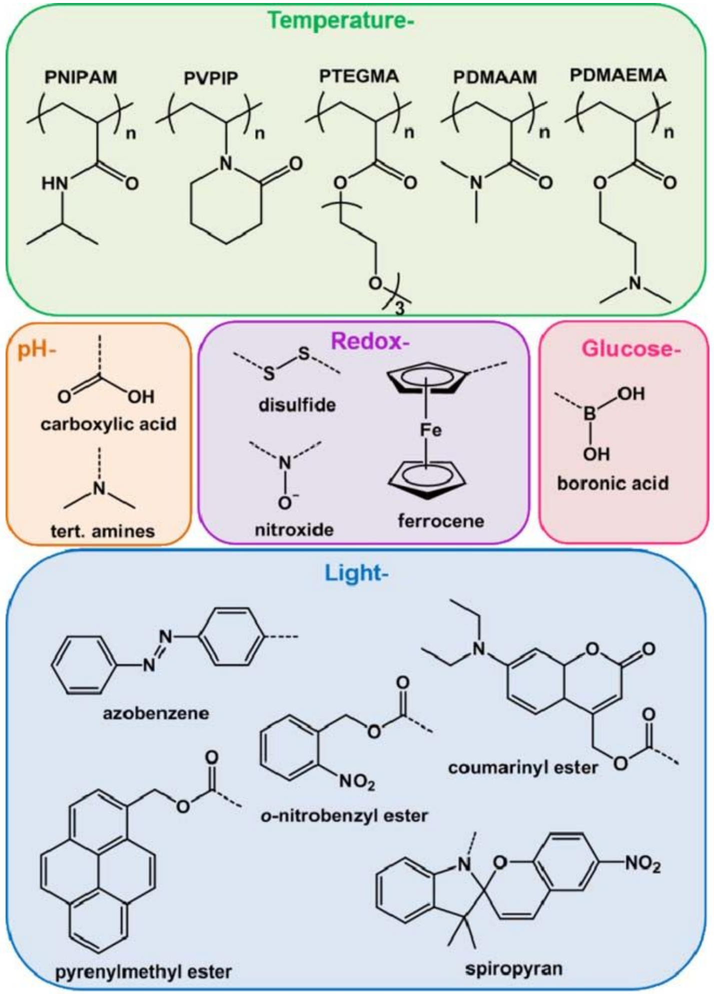
Microgels

Stimuli-Responsive Microgels

Temperature, pH, light, magnetic field, redox



Macromol. Chem. Phys. **2025**, 226, 2400472

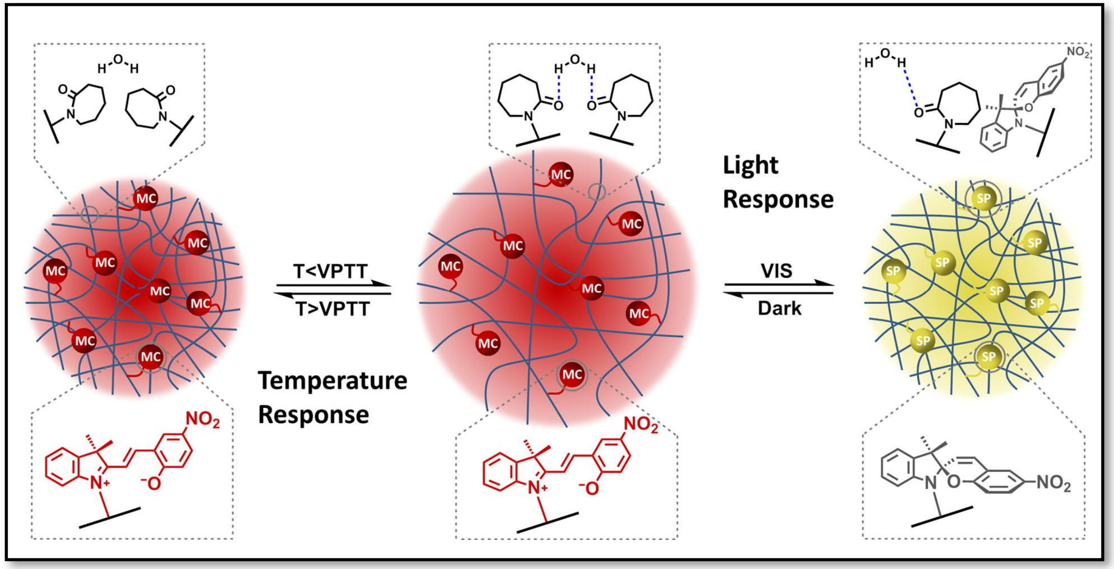


Chem. Rec. **2016**, 16, 1398

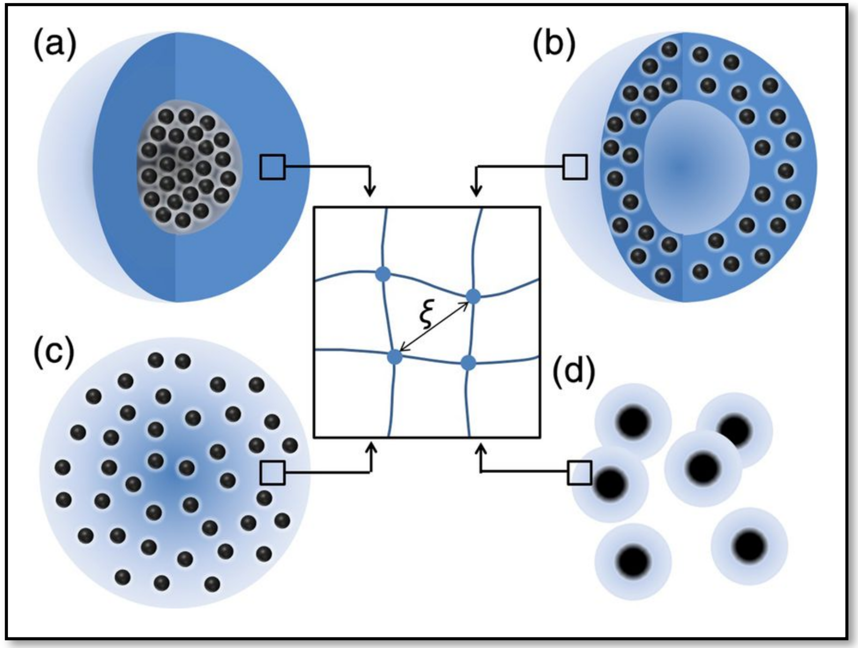
Microgels

Stimuli-Responsive Microgels

Temperature, pH, light, magnetic field, redox



J. Coll. Interf. Sci. **2021**, *582, Part B*, 1075



Bioeng Transl Med. **2021**, *6*, e10190

Microgels

Applications of Microgels

- Drug delivery (stimuli-responsive release
- Catalysis (swollen particles as nanoreactors)
- Stabilization of emulsions (Pickering-type mechanism)
- Rheology modifiers in paints, cosmetics and food formulations (softness)
- Photonic materials and chemical sensors (particle spacing changes during swelling)

… biomedical engineering, soft robotics, adaptive materials, active matter…